SingleRAN

# Performance Management Feature Parameter Description

**Issue** 03

**Date** 2025-08-30

HUAWEI TECHNOLOGIES CO., LTD.

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# Security Declaration

## Vulnerability

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# Contents



# 1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes

  Changes in functions and their corresponding parameters

- Editorial changes

  Improvements or revisions to the documentation

## 1.1 SRAN21.1 03 (2025-08-30)

This issue includes the following changes.

### Technical Changes

None

### Editorial Changes

Added the description that the cell-level performance data reported in a measurement period may be inaccurate in abnormal scenarios. For details, see **3.2.1.2 Object Type**.

Revised descriptions in this document.

## 1.2 SRAN21.1 02 (2025-06-28)

This issue includes the following changes.

### Technical Changes

None

### Editorial Changes

Revised descriptions in this document.

## 1.3 SRAN21.1 01 (2025-03-10)

This issue includes the following changes.

### Technical Changes

| Change Description | Parameter Change |
|---|---|
| Added support for flexible performance counter objects in LTE. For details, see **3.2.1.2 Object Type**. | Added parameters:<br>• **FlexCounterParamGrp**.***Index***<br>• **FlexCounterParamGrp**.***CnOperatorId***<br>• **FlexCounterParamGrp**.***Spid***<br>• **FlexCounterParamGrp**.***Qci***<br>• **FlexCounterParamGrp**.***Arp*** |
| Added support for measurement of NR DU cell QCI-level flexible performance counters in SA networking. For details, see **3.2.1.2 Object Type**. | Added the **gNBFlexCounterObjCfg**.***QciValidPolicy*** parameter.<br>Modified parameter: Added the enumerated value **NRDUCELL_FLEX** to the **gNBFlexCounterObjCfg**.***FlexCounterObjType*** parameter. |

### Editorial Changes

Revised descriptions in this document.

## 1.4 SRAN21.1 Draft A (2024-12-31)

This issue introduces the following changes to SRAN20.1 02 (2024-06-04).

## Technical Changes

| Change Description | Parameter Change |
|---|---|
| Added support for flexible performance counter objects. For details, see **3.2.1.2 Object Type**. | Added parameters:<br>• **gNBFlexCounterObjCfg**.***FlexCounterObjType***<br>• **gNBFlexCounterObjCfg**.***FlexCounterObjId***<br>• **gNBFlexCounterObjCfg**.***OperatorId***<br>• **gNBFlexCounterObjCfg**.***RfspIndex***<br>• **gNBFlexCounterObjCfg**.***Nr5qi***<br>• **gNBFlexCounterObjCfg**.***Qci***<br>• **gNBFlexCounterObjCfg**.***Arp*** |

## Editorial Changes

Revised descriptions in this document.

# 2 About This Document

## 2.1 General Statements

### Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

### Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

## 2.2 Applicable RAT

This document applies to GSM, UMTS, LTE FDD, LTE TDD, NB-IoT, and NR.

For definitions of base stations described in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview*.

## 2.3 Features in This Document

This document describes the following features.

| RAT | Feature ID | Feature Name | Chapter/Section |
|---|---|---|---|
| SRAN | MRFD-210302 | Performance Management | 3.2 Technical Description |
| LTE FDD | LBFD-004008 | Performance Management | |
| LTE TDD | TDLBFD-004008 | Performance Management | |
| NB-IoT | MLBFD-12000408 | Performance Management | |
| NR | FBFD-010025 | Basic O&M Package (specifically the performance management function) | |

# 3 Performance Management

## 3.1 Overview of Performance Management

### 3.1.1 Definition

Performance management is a function defined by the Telecommunication Management Network (TMN). This feature provides a method of efficiently monitoring network performance, while facilitating network evaluation, optimization, and troubleshooting.

### 3.1.2 Benefits

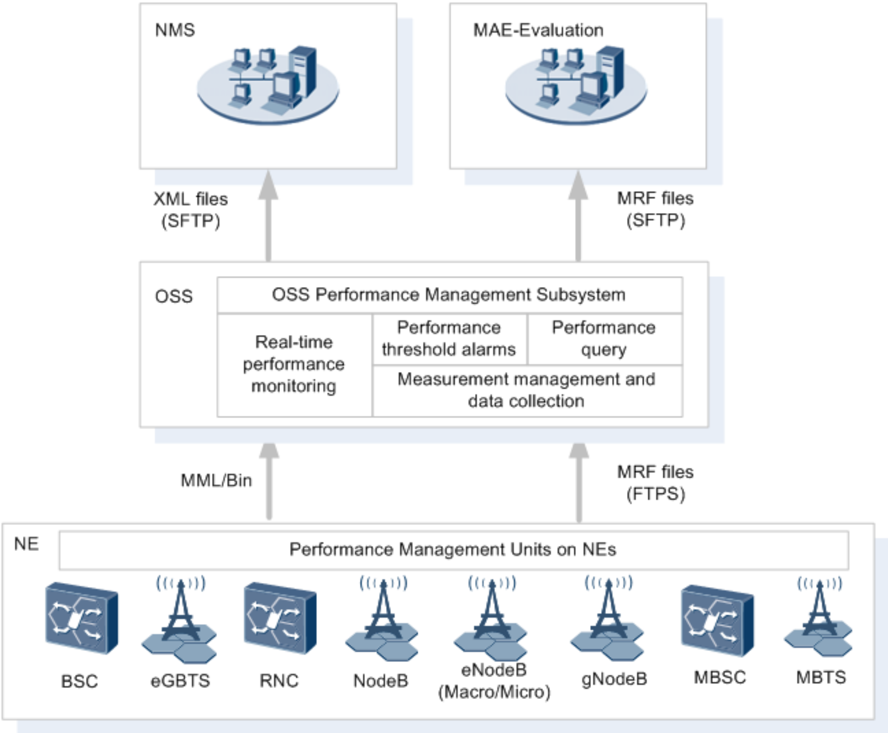
Performance management helps you:

- Check whether product-related features are effective.
- Detect networks whose performance deteriorates, enabling telecom operators to take measures to improve network quality.
- Troubleshoot network faults and provide suggestions on network quality improvement.
- Monitor and optimize wireless and transport networks. This improves user experience and helps telecom operators maximize the usage of existing device resources.
- Offer network planning engineers detailed information required for network expansion.

### 3.1.3 Architecture

Performance management includes measurement management, measurement data collection, real-time performance monitoring, and performance threshold alarms. The NEs, OSS, MAE-Evaluation, and NMS jointly provide these functions. **Figure 3-1** shows the architecture of performance management.

**Figure 3-1** Performance management architecture




| | |
|---|---|
| NMS: network management system | MAE-Evaluation: MBB Automation Engine Evaluation |
| XML: Extensible Markup Language | MRF: measurement result file |
| OSS: operations support system | MML: man-machine language |
| NE: network element | SFTP: Secure File Transfer Protocol |
| FTPS: File Transfer Protocol over SSL | |

The following describes the functions of each component in the performance management architecture.

## NEs

NEs in the performance management architecture include:

- BSC and eGBTS on the GSM network
- RNC and NodeB on the UMTS network
- Macro and other eNodeBs on the LTE network
- gNodeB on the NR network
- MBSC and multimode base station on the SingleRAN network

After receiving performance measurement result subscription from the MBB Automation Engine (MAE), NEs collect performance counters indicating the performance of physical and logical resources on the network based on the subscription information, generate corresponding performance data after each measurement period, and report the data to the MAE. After each measurement period for periodical performance measurement, the NEs store the performance data in measurement result files (MRFs) and send the MRFs to the MAE. After each measurement period for real-time performance monitoring, the NEs send measurement results in binary messages to the MAE. GBTS is not displayed in the preceding figure, because the GBTS performance statistics are generated and reported to the MAE by the BSC.

**NOTE**

If the NE reset period includes the end time of a measurement period, the NE cannot report the measurement result file. This will lead to the absence of performance counter measurement data between the start of the current measurement period and the restoration of an NE reset. When the next measurement period ends after the NE reset, the NE will normally report the measurement result file.

## MAE

The MAE is the platform to execute performance management. Users can register performance measurement tasks, and process and monitor performance data through the graphical user interface (GUI). The MAE provides the following performance management functions:

- Performance data collection and storage

  The MAE periodically obtains measurement results of performance counters from NEs based on measurement settings and saves them to the MAE database. Users can modify the measurement settings and save them as a template. Users can also view measurement status, synchronize measurement data, and view synchronization task status.

- Performance data query

  On the MAE, users can query performance data saved in the database. The MAE displays the query results in a table, line chart, or bar chart. The MAE also allows users to print and save query results, to query templates and missing data, and to subscribe to performance data.

- Performance threshold alarms

  Users can set alarm thresholds for measurement counters on the MAE. When a counter value exceeds a specified threshold, a performance threshold alarm is generated. When the counter value falls below the threshold, the performance threshold alarm is cleared automatically.

- Performance data export

  The MAE performance management provides source data for the NMS and MAE-Evaluation. For details of the MAE performance management subsystems, see the OSS document *Performance Report Management*.

## MAE-Evaluation and NMS

After obtaining the performance data from the MAE, the MAE-Evaluation parses and stores the data in the database and regularly aggregates the data. The MAE-Evaluation can store key data for an extended period, query data flexibly in multiple dimensions, and manage, generate, and distribute reports.

After NEs send performance MRFs to the MAE, the NMS can obtain the files from specified directories on the MAE.

For details of the MAE-Evaluation, see *RAN Statistics Performance Visibility* in *MAE-Evaluation Product Documentation*.

# 3.2 Technical Description

## 3.2.1 Basic Concepts

Three measurement elements are mandatory for periodical performance measurement and real-time performance monitoring: performance counters, object types, and measurement periods. Function subsets and function sets are introduced to implement layered performance counter measurement.

### 3.2.1.1 Performance Counter

Performance counters, also termed as counters, indicate the number or amount (such as service request times and traffic) collected for a measurement purpose.

Each performance counter has a unique ID and name. **Table 3-1** describes an example performance counter.

**Table 3-1** Performance counter example

| ID | Name | Description |
| --- | --- | --- |
| 67180678 | **VS.AMR.DL.RateUp** | Times of increasing the downlink rate for adaptive multirate (AMR) voice services in a cell |

Performance counters are classified into class A counters, class B counters, and user-defined performance counters.

#### Class A Counters

Class A counters refer to counters in the traditional performance architecture and include common and extended performance counters.

- Common performance counters, also known as raw counters, are the key counters defined by NEs. Measurements for these counters start by default. If you do not clear these counters, all the measurement results of these counters are reported to the OSS and stored in the OSS performance database for you to query. You can clear common performance counters.
- Extended performance counters are defined by NEs. Measurements for these counters do not start by default. You can select or deselect extended performance counters when the OSS is running. If you select such a counter, the measurement results of this counter are saved in the OSS performance database for you to query. If such a counter is not selected, the measurement

results of this counter are not saved in the OSS performance database. As a result, you cannot query the measurement results of this counter.

## Class B Counters

Class B counters refer to the counters in the new performance architecture. The new performance architecture is introduced to increase the number of performance counters supported by the base station OM resources and reduce the base station resource consumption.

Class B counters are defined by NEs. Measurements for these counters do not start by default. You can activate or deactivate class B performance counters when the OSS is running. If you activate such a counter, the measurement results of this counter reported by NEs are saved in the OSS performance database for you to query. If such a counter is not activated, the measurement results of this counter are not reported to the OSS by NEs. As a result, you cannot query the measurement results of this counter. Only the gNodeB supports class B counters.

## User-defined Performance Counters

User-defined performance counters are customized by users on the OSS client. Performance counters are customized using the following methods:

Method 1: Perform arithmetic operations (add, subtract, multiply, and divide) on common and extended performance counters. User-defined performance counters include the KPIs defined by Huawei and by operators. For details about the KPIs recommended by Huawei, see *KPI Reference* in Huawei ICS Lite documentation package. The KPIs described in *KPI Reference* are based on Huawei experience in commercial network management and radio network optimization. You can modify these KPIs or define new KPIs on the OSS as required.

Method 2: Add filter criteria to base counters. Counters that are defined by using this method are also called event-based counters (EBCs). The common types of EBCs are as follows: accessibility, retainability, mobility, delay, and cell resource. Each type includes multiple base counters. For example, **Number of RRC Connection Establishment Attempts (Excluding Duplicated Attempts)** is a base counter. An EBC measuring the number of RRC connection establishment attempts in emergency calls is created by setting the **Establishment Cause** filter criterion to **RRC_EMERGENCY** for this base counter. The default measurement periods of EBCs do not map the supported measurement periods of base counters. Currently, EBCs support only the 15-minute measurement period while base counters support short and long measurement periods. For details about the short and long measurement periods, see **3.2.1.5 Measurement Period**. EBC statistics are collected on the OSS after call events are filtered by the access cause, failure cause, service characteristic, and operator ID.

NOTE

- After you start a measurement task for user-defined performance counters on the OSS, the OSS automatically enables the measurement of common performance counters and extended performance counters involved in the user-defined performance counters.
- Class A and class B counters cannot be defined in the same user-defined performance counter.
- eNodeBs support EBCs as of eRAN3.0. RNCs support EBCs as of RAN15.0.
- A gNodeB does not support EBCs.

### 3.2.1.2 Object Type

An object type indicates a physical or logical entity that is measured, such as a base station, an Ethernet port, and an RRC connection. Performance counters are measured for an object.

Object types are classified as either deterministic or custom.

## Deterministic Object Type

The physical or logical entity measured for a deterministic object is fixed. **Table 3-2** lists the example deterministic object types for some NodeB performance counters.

**Table 3-2** Examples of deterministic object types

| Counter ID | Counter Name | Description | Object Type |
|---|---|---|---|
| 50333220 | VS.BTS.EnergyCons.Adding | Total power consumption of all boards for a NodeB | NodeB |
| 1542455297 | VS.FEGE.TxBytes | Number of bytes in the packets successfully transmitted on the Ethernet port | ETHPORT |
| 1526726657 | L.RRC.ConnReq.Msg | Number of RRC connection setup requests (retransmission included) | Cell |
| 1526728251 | L.E-RAB.SuccEst.PLMN | Total number of successful E-RAB setups initiated by UEs for a specific operator in a cell | Cell.Operator<br>**NOTE**<br>The operator object is co-defined by the cell and operator attributes. |

| Counter ID | Counter Name | Description | Object Type |
|---|---|---|---|
| 1911827186 | N.QosFlow.Est.Att.Cell5QI.EPSFB | Number of EPS fallback-triggered QoS flow setup attempts for a certain 5QI in a cell | NRCELL.5QI<br>**NOTE**<br>The cell 5QI object is co-defined by the cell and 5QI attributes. |

**NOTICE**

In abnormal scenarios such as cell deactivation, the cell-level performance data reported in a measurement period may be inaccurate.

## Custom Object Type

To improve the flexibility of object deployment, flexible object performance measurement is introduced, as described in **Table 3-3**.

**Table 3-3** Examples of custom object types

| Counter ID | Counter Name | Description | Object Type |
|---|---|---|---|
| 1911849336 | N.NsaDc.SgNB.Add.Att.Flex | Number of SgNB addition attempts for a flexible object in LTE-NR NSA DC scenarios | NRCell.Flex |
| 1911849322 | N.ThpVol.DL.LastSlot.Flex | Downlink RLC-layer traffic volume in the last slots during which the buffer becomes empty for a flexible object | NRDUCell.Flex |
| 1526763125 | L.Thrp.bits.DL.FlexQci.Index0 | Total traffic volume of transmitted downlink PDCP-layer data for services with a flexible QCI of counter index 0 | FlexQci.Cell |

The **gNBFlexCounterObjCfg**.***FlexCounterObjType*** parameter can be set to **NRCELL_FLEX_AND_NRDUCELL_FLEX** or **NRDUCELL_FLEX** to use the measurement of cell flexible object performance (NRCell.Flex) and measurement of DU cell flexible object performance (NRDUCell.Flex). The parameters listed in **Table 3-4** can be configured to map one or more objects to a flexible performance counter object ID specified by the **gNBFlexCounterObjCfg**.***FlexCounterObjId*** parameter.

**Table 3-4** Dimensions for flexible performance counter measurement in NR

| Dimension | Parameter |
|---|---|
| Operator | **gNBFlexCounterObjCfg**.***OperatorId*** |
| RAT/Frequency Selection Priority | **gNBFlexCounterObjCfg**.***RfspIndex*** |
| NR 5G QoS Identifier | **gNBFlexCounterObjCfg**.***Nr5qi*** |
| QoS Class Identifier | **gNBFlexCounterObjCfg**.***Qci*** |
| Allocation and Retention Priority | **gNBFlexCounterObjCfg**.***Arp*** |

NOTICE

In NR, the configuration of objects for flexible performance counter measurement can be dynamically modified online. After the modification, the base station may report the values of both the original and new measurement counters within a period of time, causing inaccurate counter values in the current and next measurement periods.

The **FLEX_QCI_MEAS_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter or the **FDD_HVC_EXP_GUARANTEE_SW** or **TDD_HVC_EXP_GUARANTEE_SW** option of the **EnodebAlgoExtSwitch**.***NetworkPrfmOptSwitch*** parameter can be selected to use flexible QCI measurement (FlexQci.Cell). The parameters listed in **Table 3-5**, which determine the dimensions for flexible performance counter measurement, can be configured to map one or multiple objects to a flexible performance counter object index specified by the **FlexCounterParamGrp**.***Index*** parameter. The operator dimension is optional, and other dimensions are mandatory.

**Table 3-5** Dimensions for flexible performance counter measurement in LTE

| Dimension | Parameter |
|---|---|
| Operator | **FlexCounterParamGrp**.***CnOperatorId*** |
| SPID | **FlexCounterParamGrp**.***Spid*** |
| QoS Class Identifier | **FlexCounterParamGrp**.***Qci*** |
| Allocation and Retention Priority | **FlexCounterParamGrp**.***Arp*** |

NOTICE

In LTE, the configuration of objects for flexible performance counter measurement can be dynamically modified online. The modification takes effect immediately for UEs that newly access the network but does not take effect immediately for UEs that have been online before the modification.

For performance counter measurement related to flexible QCIs, the operator, SPID, QCI, and ARP in the flexible counter parameter group specified by the **FlexCounterParamGrp**.***Index*** parameter take effect. For performance counter measurement related to UEs with flexible attributes, only the SPID in the flexible counter parameter group specified by the **FlexCounterParamGrp**.***Index*** parameter takes effect, and the settings of other dimensions are invalid.

### 3.2.1.3 Function Subset

Function subsets are introduced to implement layered performance counter management. Function subset indicates the effective combination of counters with the same attributes. For example, hard handover measurement is a function subset. It is used to calculate the various messages in the basic hard handover procedure from the perspective of a cell.

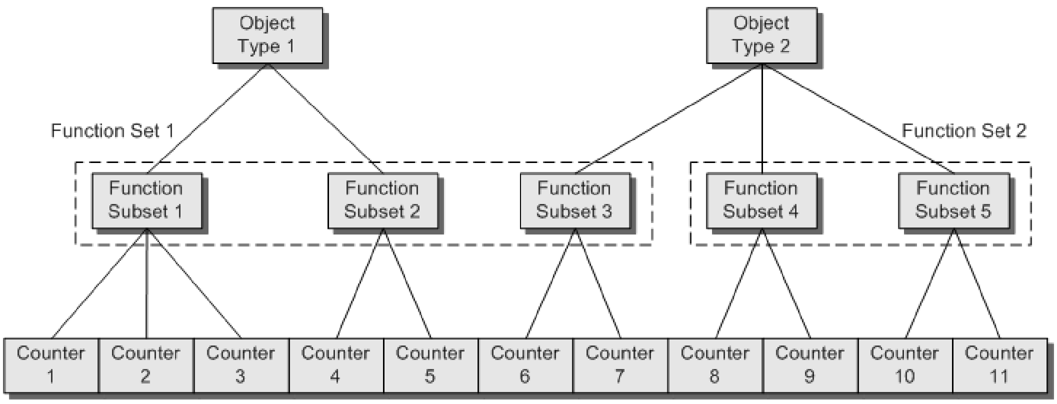
### 3.2.1.4 Function Set

A function set indicates a set of several function subsets with the same attributes. For example, the function set **Measurements related to Radio Access Bearer (RAB) management** consists of the following function subsets: **Measurement of CS RAB assignment modification per RNC**, **Measurement of CS RAB release per cell**, and **Measurement of PS RAB assignment setup per RNC**.

In the MAE, multiple counters with the same measurement purpose form a function subset, and multiple function subsets with the same attribute form a function set. With these concepts, you can perform performance measurement more conveniently and efficiently, and easily locate the required counter.

A performance counter belongs to only one function subset, and a function subset belongs to only one function set. Figure 3-2 shows the relationships between object types, function sets, function subsets, and performance counters.

**Figure 3-2** Relationships between object types, function sets, function subsets, and performance counters



### 3.2.1.5 Measurement Period

NEs report measurement results to the MAE based on specified measurement periods. The MAE analyzes the results and then saves the results to the database. Therefore, you can query the measurement result of the required period. The periods for reporting measurement results may vary according to performance counters and NE types. **Table 3-6** lists the measurement periods supported by NEs.

**Table 3-6** Measurement periods supported by NEs

| RAT | NE Type | Supported Measurement Period (Minute) | Default Measurement Period (Minute) |
|---|---|---|---|
| GSM | BSC6000 GSM | 15, 60, and 1440 | 60 |
| | BSC6900 GSM<br>BSC6910 GSM | 15, 60, and 1440 | 60 |
| | eGBTS | 15 and 60 | 60 |
| GSM | BSC6960 GSM | 15 and 60 | 60 |
| UMTS | BSC6810 UMTS | 5, 30, and 1440 | 30 |
| | BSC6900 UMTS<br>BSC6910 UMTS | 5, 30, and 1440 | 30 |
| | NodeB | 15 and 60 | 60 |
| UMTS | BSC6960 UMTS | 15 and 60 | 60 |
| LTE | eNodeB | 15 and 60 | 60 |
| NR | gNodeB | 15 and 60 | 60 |
| SingleRAN | BSC6900 GU<br>BSC6910 GU | 15, 60, and 1440 | 60 |
| | 3900 & 5900 series multimode base stations | 15 and 60 | 60 |

**NOTE**

Measurement periods are classified into short measurement periods and long measurement periods. A short measurement period can be 5 minutes or 15 minutes, and only one period can be set at a time. A long measurement period can be 15 minutes, 30 minutes, or 60 minutes, and only one period can be set at a time. The short and long measurement periods cannot be set to 15 minutes concurrently.

For each NE, the number of 5-minute period counters that can be measured at the same time must not exceed 1000. This helps save storage space of the MAE.

# 3.2.2 Performance Counter Statistical Methods

## 3.2.2.1 Performance Counter Statistical Types

**Table 3-7** describes the performance counter statistical types.

**Table 3-7** Performance counter statistical types

| Statistical Type | Description | Example |
| --- | --- | --- |
| SCAN | Counts the number of sampling times in a statistical period. The accumulated sample value increases by 1 at each time when the number of sampling times is increased by 1. The initial value is **0**. | **Sampling Times of PDCH Measurement**<br>If the sampling interval is 5 seconds and statistical period is 30 minutes, the value of this counter is **360**. |
| ACC | Obtains accumulated sample values in a statistical period. The average counter value is obtained by dividing the accumulated sample values by the number of sampling times. | **Total Number of Sampled Available PDCHs**<br>The value of this counter is the total number of PDCHs that are sampled every five seconds.<br>You can also obtain the average counter value in 30 minutes. |
| GAUGE | Counts the variable whose value can be dynamically increased or decreased. The variable is of **int** or **float** type. After the statistical period ends, the variable value is the statistical result. | **Number of UEs in CELL_DCH State for Cell** |
| PEG | The number of events is increased by 1 each time. | **Number of Incoming Hard Handover Attempts for Cell** |
| PDF | Perform distribution statistics based on intervals. | For **Number of Times that DL Throughput of HSDPA Services Within Each Range**, interval distribution for statistics is as follows:<br>[0]: 0 <= x < 32 kbit/s [1]: 32<= x < 64 kbit/s [2]: 64 <= x < 256 kbit/s [3]: 256<= x < 512 kbit/s ... [13]: 12288 kbit/s <= x |

#### 3.2.2.2 Performance Counter Aggregation Types

Performance counter aggregation types indicate how to aggregate counter values by time or by object. **Table 3-8** lists the aggregation types.

**Table 3-8** Performance counter aggregation types

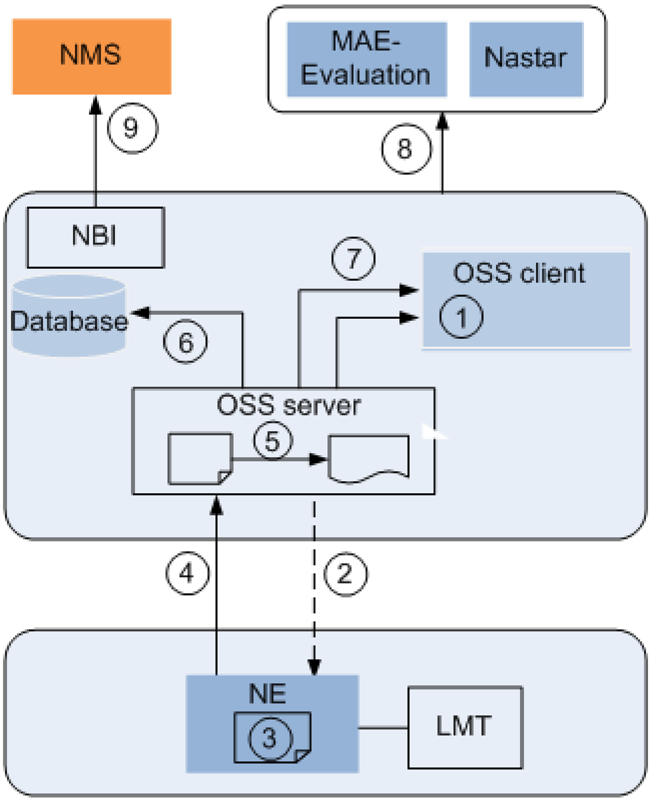
| Aggregation Type | Description | Example |
| --- | --- | --- |
| SUM | Sum of all samples | Number of bytes in the IP packets received on the SCTP |
| MAX | Maximum value of all samples | Maximum IP packet receive rate on the SCTP |
| MIN | Minimum value of all samples | Minimum IP packet receive rate on the SCTP |
| AVG | Average value of all samples | Average IP packet receive rate on the SCTP |

### 3.2.3 Counter-based Performance Measurement Process

After you set the performance counters, object types, and measurement periods for an NE on the MAE, the NE reports the counter measurement results (performance data) to the MAE when the measurement period begins. The mediation service of the MAE server parses the packets reported by NEs and converts them into data of a uniform format. The performance service saves the data required by users to the performance database. Through the northbound interface of the MAE, the NE performance data is sent to the NMS for northbound users to analyze.

**Figure 3-3** shows the performance measurement process.

**Figure 3-3** Counter-based performance measurement process

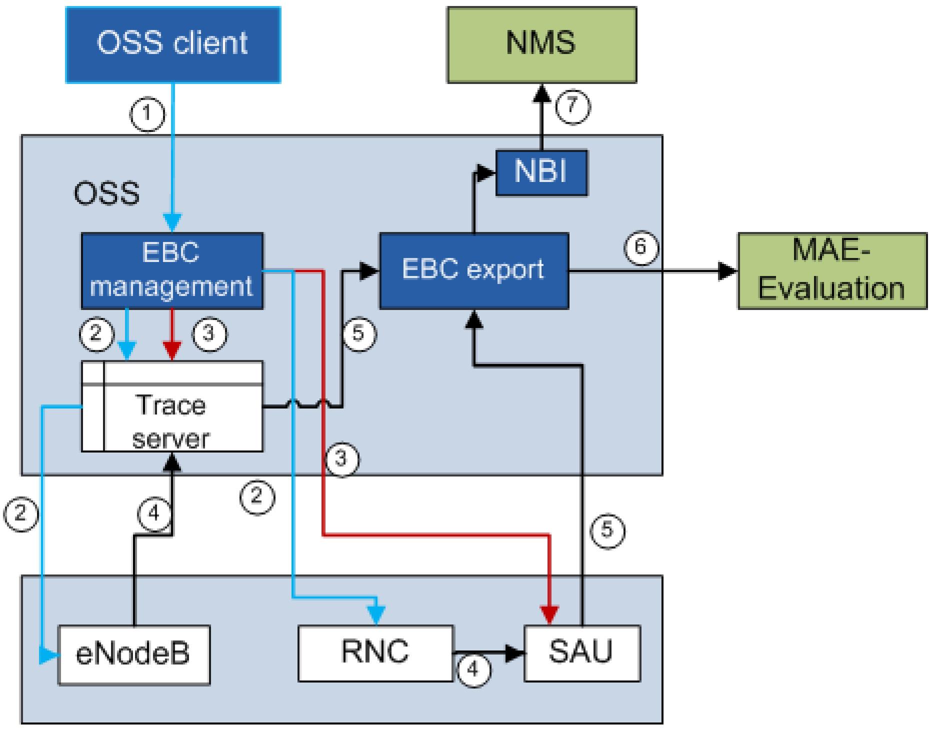



The following table describes the basic process of counter-based MAE performance measurement.

| No. | Procedure | Description |
| --- | --- | --- |
| 1 | Register measurement tasks | On the MAE client, set the measurement counters, object types, and period, and deliver measurement tasks to the performance management (PM) mediation through the PM server. |
| 2 | Deliver measurement tasks | The PM mediation delivers the measurement counters, object types, and period to NEs. |
| 3 | Generate measurement results | The NEs generate the performance measurement results based on the measurement period. |
| 4 | Report measurement results | The NEs report the performance measurement results to the MAE server.<br>The protocol used by an NE to report the performance measurement results is determined by NE attributes. Different NEs may use different protocols to report performance measurement results. |

| No. | Procedure | Description |
|---|---|---|
| 5 | Parse measurement results | The PM mediation parses the performance measurement results. |
| 6 | Import measurement results | The MAE server saves the parsed performance measurement results to the performance database. |
| 7 | Query results on the MAE client | Users query reported performance measurement results through the MAE client. |
| 8 | Export performance data | The performance data in the database is provided to the NMS through the MAE northbound interface.<br>The Nastar uses a data collection tool to obtain performance data based on the IP address of the MAE server, user name, password, and the path for saving performance data.<br>The MAE-Evaluation allows you to use the task management function to collect data from the MAE, save the data in a file on the MAE-Evaluation, and import data in the file to the database as scheduled for subsequent query and analysis. |

**Figure 3-4** EBC-based performance measurement process



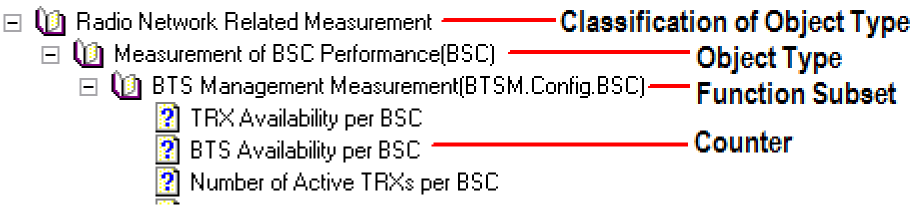
The following table describes the basic process of EBC-based MAE performance measurement.

| No. | Procedure | Description |
| --- | --- | --- |
| 1 | Register measurement tasks | Users set and activate measurement counters on the MAE client and deliver the counters to the EBC management module through the performance server. |
| 2 | Deliver CHR event measurement tasks | The EBC management module converts activated measurement counters into CHR events that need to be subscribed to, and delivers the events to eNodeBs (through the Trace Server) and RNCs. |
| 3 | Deliver counter-defined script files | The EBC management module delivers counter-defined script files to the Trace Server and SAU for counter pre-calculation. |
| 4 | Report CHR files | eNodeBs and RNCs report CHR files to which the EBC subscribes to the Trace Server and SAU. |
| 4 | Calculate counters | The Trace Server and SAU use the counter-defined script files to process CHR files and generate counter statistical results. |
| 5 | Generate result files | The Trace Server and SAU report counter pre-calculation results to the EBC export module which then generates counter result files. |
| 6 | Query counters | The MAE-Evaluation collects counter result files generated by the EBC export module as scheduled, parses the files, and saves the files to the database for you to query on the client. |
| 7 | Export performance data | The northbound module converts the counter results files generated by the EBC export module into northbound counter result files in XML format for the NMS to use. |

## 3.2.4 Performance Counter Help Documents

Detailed information about performance indicator definitions and calculation formulas of the NEs are described in the performance counter reference released with the NE software. Content in performance counter reference is organized by object type. **Figure 3-5** uses *BSC6900 GSM Performance Counter Reference* as an example.

**Figure 3-5** Content organization



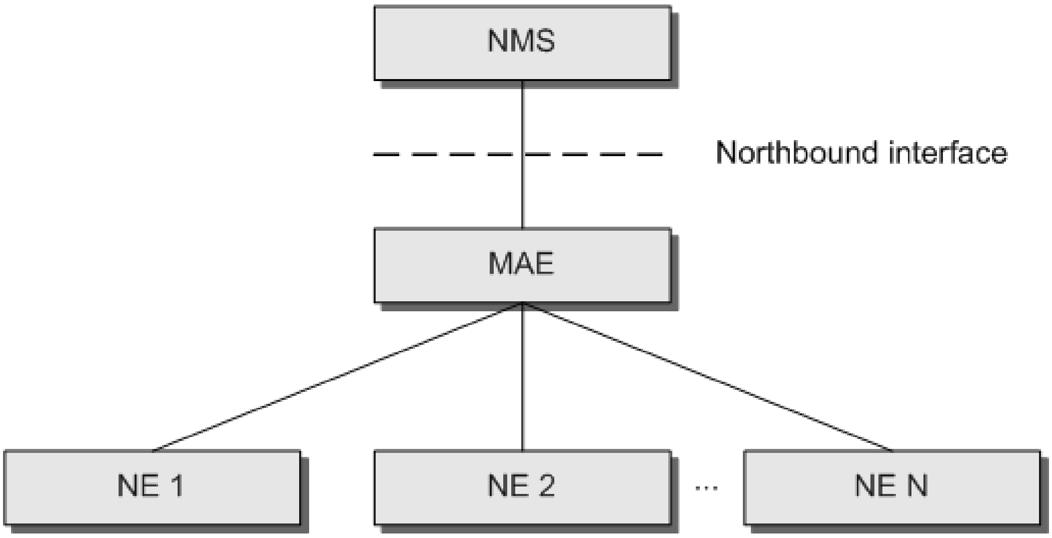

  

# 3.2.5 Northbound Interface

## 3.2.5.1 Northbound Interface Overview

The northbound interface is an interface between the Element Management System (EMS) and the NMS, as shown in **Figure 3-6**. The MAE in the following figure is the EMS.

**Figure 3-6** Position of the northbound interface




The MAE opens the northbound interface for performance files to support the capability of integrating with the NMS manufactured by a third party. It is recommended that the northbound interface for performance files adopt the XML format, which is defined by 3GPP protocols for performance measurement result files. File export delay through this interface is short and the MAE is least affected.

The file interface can periodically export performance data from the database to files based on preset export criteria. The NMS can obtain the exported performance files from a specified path on the MAE by using the SFTP. If the NMS provides the IP address of a server and specifies a path on the server, the MAE can upload files to the path on the server.

For details of the northbound performance file interface, see *iMaster MAE-Access Northbound Performance File Interface Developer Guide (NE-based)*. To obtain the document, contact Huawei engineers.

## 3.2.5.2 Recommended Interconnection Solution

For Huawei NEs, a counter ID uniquely identifies a counter and does not change according to NE versions.

If counter IDs meet the interconnection requirements, you are advised to interconnect the NMS with the MAE based on counter IDs to reduce the impact of counter name difference on the NMS.

Counter names are longer that counter IDs and may vary based on NE versions.

NOTE

It is not recommended that counter names be used to interconnect the MAE with the NMS, as the lengthiness of counter names result in large-sized northbound performance files, which affects file transfer and parsing efficiency. In addition, the counter names may change occasionally.

#### 3.2.5.3 Related Documents

When interconnecting the NMS manufactured by a third party through the northbound interface, you can refer to the following documents:

- *MAE Northbound Performance File Interface Developer Guide (NE-Based)*

  This document is released with the MAE software. This document is intended to provide the guidance on how to interconnect the NMS manufactured by a third party with the MAE through the northbound performance file interface. This document includes the mechanism of the performance file interface, performance file format, and user commissioning guide.
- *Performance Counter List*

  This document is released with the NE mediation. It includes the definition of counters supported by the NE of the relevant version and the format of counter objects.
- *NE Performance Counter Changes*

  This document is released with the NE software. It includes the counter change between the current NE version and earlier versions. You must modify the counters in the NMS according to this document before NMS upgrade. For any counter change problem, contact the onsite engineers.

## 3.2.6 Schemes for Adding or Deleting Performance Counters and Object Type Parameters

In normal cases, performance counters are added in a new version for the following reasons:

- New features are introduced.
- Original features and statistical processes are optimized.

Performance counters may be deleted for the following reasons:

- The performance counters are no longer used (related algorithms are changed or the performance counter design is optimized) or will be replaced by other performance counters due to feature changes.
- Object types for the performance counters are deleted due to hardware, function, or feature changes.

In addition, some parameters of an object type may be deleted or replaced by other parameters after configuration model optimization. Adding or deleting performance counters and object type parameters has impact on northbound interface interconnection.

Therefore, performance counters and object type parameters to be deleted will be reserved for two NE versions: the current version and the later version. **Table 3-9** describes the schemes for deleting performance counters and object type parameters.

**Table 3-9** Schemes for deleting performance counters and object type parameters

| Scenario | Scheme (N-N+1) | Scheme (N+2) |
|---|---|---|
| Old performance counters are replaced by new performance counters. | • Old performance counters are reserved, and statistics about these performance counters are reported properly or are null values.<br>• Statistics about new performance counters are reported properly. | Old performance counters are deleted. |
| Old performance counters are deleted. | Old performance counters are reserved, and statistics about these performance counters are reported properly or are null values. | Old performance counters are deleted. |
| Old object type parameters are deleted or replaced by other parameters due to configuration model optimization. | • Old object type parameters are reserved and reported. The parameter value validity depends on the validity of values for the corresponding parameters in the configuration model.<br>• New object type parameters are reported properly. | Old performance counters are deleted. |

Table 3-10 lists the documents containing information about deleted performance counters and object type parameters.

**Table 3-10** Documents containing information about disused performance counters and object type parameters

| Document Name | Description |
|---|---|
| Disuse Performance Counter List | This document is released with the NE software and describes all deleted counters, and counters and object types to be deleted in this version. It also provides disuse statements that describe the schemes and reasons for deleting these counters or object types. |
| Performance Counter Reference | Disuse statements describing the schemes for deleting performance counters are provided in the **Description** field. This document is included in the NE ICS Lite documentation package. |

# 4 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

# 5 Reference Documents

1. *BSC6900 GSM Performance Counter Reference*
2. *BSC6910 GSM Performance Counter Reference*
3. *BSC6960 GSM Performance Counter Reference*
4. *GSM KPI Reference*
5. *BSC6900 UMTS Performance Counter Reference*
6. *BSC6910 UMTS Performance Counter Reference*
7. *BSC6960 UMTS Performance Counter Reference*
8. *WCDMA KPI Reference*
9. *eNodeB KPI Reference*
10. 3900 & 5900 Series Base Station Performance Counter Reference
11. *MAE Performance Measurement Management User Guide*
12. *MAE Performance Report Management User Guide*
13. *iMaster MAE-Access Northbound Performance File Interface Developer Guide (NE-Based)*